

1: Tell Us About your self:

Hello sir/mam. My name is Vishal Soni, and I'm currently pursuing my Bachelor of Engineering in Information Technology from IIIT Pune.

I'm a systems-focused developer with a strong interest in backend engineering, workflow-driven systems, automation, and solving operational problems through technology.

I'm currently working as a Full Stack Developer Intern at Attento Technologies, where I work on scalable backend systems, workflow-driven admin platforms, and automation-related features using technologies like Next.js, NestJS, TypeScript, and Firebase.

During my internship, I've worked on API development, debugging, documentation, and understanding how different workflows and backend processes connect together to build reliable operational systems.

While preparing for this opportunity, I started learning more about MES and manufacturing workflows, and I found the domain very interesting because it combines systems, operations, process management, and technology together.

I believe my technical background, structured problem-solving approach, willingness to learn, and interest in workflow-oriented systems make me a strong fit for this role.

2: Walk us through your internship at Attento Technologies:

During my internship at Attento Technologies, I worked as a Full Stack Developer Intern and contributed to multiple internal operational platforms and automation-focused systems.

One of the major projects I worked on was an Admin Console, which was a workflow-driven full stack application used for managing internal resources, client entities, and operational mappings required to serve customers efficiently.

My role involved developing backend APIs, handling entity relationships, implementing business logic, and building frontend modules using technologies like Next.js, NestJS, TypeScript, and MongoDB. Through this project, I gained experience in designing structured workflows, managing interconnected systems, and understanding how operational processes are handled in real-world environments.

I also worked on Firebase Cloud Functions where I implemented scheduled jobs and cron-based automations to streamline business workflows and reduce manual operational effort. This helped me understand event-driven systems and backend automation processes.

Another interesting project was a Road Feature Analyzer, where we used MongoDB geospatial queries to visually analyze road-related data captured by internal products. My

work involved backend data processing, geospatial querying, and helping create meaningful visual representations of the data.

I also contributed to a Trip Analyzer platform, which was a full stack application that visually analyzed user trip data using Google Maps APIs. I worked on integrating APIs, handling trip-related data processing, and building visualization-oriented features.

Overall, my internship gave me strong exposure to backend systems, workflow-driven applications, automation, debugging, documentation, and collaborative development in production-oriented environments.

3: You mention 'documenting workflows and feature requirements' in your internship — can you give a specific example?

Yes. During my internship, one of my responsibilities was documenting workflows, backend functions, and API behavior in a structured way so that other developers and team members could easily understand and maintain the system.

One specific example was documenting Firebase Cloud Functions that were used for workflow automation and scheduled business operations.

I created detailed documentation in ClickUp where I systematically described each function using sections like function name, purpose, trigger type, authentication and security considerations, collections queried, input arguments, validations, expected outputs, error handling, logging behavior, test cases, and client-side expectations.

The goal was to make the workflow easy to understand for both current and future team members, especially because many of these functions were interconnected with operational processes and automated tasks.

Through this experience, I realized that good documentation is very important in workflow-oriented systems because it improves collaboration, troubleshooting, onboarding, and long-term maintainability of the platform.

4: Discuss your Admin Console SaaS Dashboard project — what challenges did it address, and how was the system designed?

The Admin Console was a workflow-driven internal SaaS platform designed to help manage client resources, operational entities, and internal business processes in a centralized system.

The main problem it solved was reducing manual coordination by organizing different operational workflows, mappings, and resource management activities into a structured dashboard.

From an architecture perspective, it was a full stack application built using Next.js on the frontend and NestJS on the backend, with MongoDB as the database.

My role involved developing backend APIs, handling entity relationships and validations, implementing business logic, and contributing to frontend modules and workflow-oriented features.

One of the key things I learned from this project was how interconnected workflows, structured data management, and clear backend architecture help build reliable operational systems.

5. What is MES (Manufacturing Execution System)? What does it do in a factory or pharmaceutical plant?

“MES stands for Manufacturing Execution System. It is a software system used to monitor, track, document, and manage production activities in real time on the manufacturing floor.

It acts as a bridge between ERP systems, which handle business planning and inventory, and the actual shop-floor operations where production happens.

MES helps manage things like work orders, batch records, operator instructions, equipment status, production tracking, and quality checks.

In pharmaceutical manufacturing, MES is especially important because it helps standardize workflows, reduce manual errors, improve traceability, and maintain proper production documentation and compliance.”

6. Why is MES especially important in the pharmaceutical industry?

MES is very important in the pharmaceutical industry because pharma manufacturing is highly regulated and directly impacts patient safety.

Pharmaceutical companies must maintain strict compliance with standards like GMP and regulations such as FDA.

MES helps by providing electronic batch records, production traceability, audit trails, quality monitoring, and standardized operational workflows.

It also reduces manual paperwork and minimizes the risk of human errors in processes like mixing, dosing, packaging, and quality verification.

Overall, MES improves compliance, reliability, traceability, and operational efficiency in pharmaceutical manufacturing.”

7. Briefly explain the ISA-95 hierarchy architecture and where Enable3 positions itself.

ISA-95 is an industrial automation standard that defines how enterprise business systems and manufacturing systems should interact in a factory environment.

The hierarchy is generally divided into levels:

Level 0 → Physical production process

Level 1 → Sensors and machines

Level 2 → Monitoring and control systems like SCADA/PLC

Level 3 → MES systems managing production workflows and operations

Level 4 → ERP systems handling business planning and enterprise operations

Enable3 mainly operates around Level 3, where MES systems connect shop-floor operations with enterprise-level planning systems.

Their role involves workflow management, manufacturing process execution, data tracking, compliance, and operational integration within pharmaceutical manufacturing environments.”

8. Why are you interested in a functional consultant role rather than a pure development role?

“I enjoy software development, especially backend and workflow-oriented systems, but over time I realized that I’m more interested in understanding how technology solves real operational and business problems rather than focusing only on coding isolated features.

During my internship, I particularly enjoyed understanding workflows, entity relationships, process logic, documentation, and how different systems interact together operationally.

That is one of the reasons this functional consultant role interests me. It combines technical understanding with process analysis, workflow optimization, communication, and real-world problem solving.

I also find the MES and pharmaceutical domain interesting because it involves structured systems, automation, compliance, and operational efficiency, which aligns well with my interests and learning mindset.”

9. What is a process map, and why is it important in MES implementation projects?

“A process map is a structured visual representation of how a business or operational workflow functions step by step, including activities, decisions, inputs, outputs, and responsible users or systems.

In MES implementation projects, process maps are important because they help understand how manufacturing operations currently work and how they should function after system implementation.

They help identify workflow dependencies, inefficiencies, validations, and operational requirements before configuring or implementing MES systems.

From my internship experience working on workflow-driven systems and documentation, I understand how important structured process understanding is for building reliable operational platforms.”

TECHNICAL QUESTIONS

1. What is NestJS and why did you choose it over Express.js for your projects?

“NestJS is a scalable backend framework built on top of Node.js and Express/Fastify. It provides a structured architecture using modules, decorators, controllers, services, and dependency injection.

I chose NestJS over plain Express.js mainly because of maintainability and scalability. In larger workflow-driven systems like the Admin Console project I worked on during my internship, having a modular and structured architecture becomes very important.

NestJS also has built-in support for validation, authentication, Swagger documentation, and dependency injection, which helped us develop enterprise-style backend systems more efficiently.

Express is very flexible and lightweight, but NestJS provides better organization for larger applications involving multiple workflows, entities, and business logic.”

2. How does Socket.io work in your Together chat app, and what challenges did you face with real-time sync?

“In my Together chat application, Socket.io was used to enable real-time bidirectional communication between clients and the server.

After the initial HTTP connection, Socket.io establishes a persistent WebSocket connection, allowing events like sending messages, typing indicators, and online status updates to happen instantly without repeated API polling.

On the frontend, I used Zustand for managing message and socket-related state efficiently.

One challenge I faced was handling synchronization issues during reconnects or rapid message updates. Sometimes messages could appear duplicated or out of order during unstable connections.

To handle this, I improved event handling logic, maintained proper state synchronization, and ensured message updates were processed consistently on the client side.

This project helped me understand event-driven systems and real-time communication workflows much better.”

3. You list TypeScript as a primary language — what advantages does it give over plain JavaScript?

“TypeScript adds static typing on top of JavaScript, which helps catch errors during development instead of at runtime.

In larger applications involving multiple APIs, entities, and workflows, TypeScript improves maintainability, readability, and developer productivity through better IDE support and type safety.

During my internship, TypeScript was especially useful while handling complex backend request and response structures in NestJS APIs. Strong typing helped identify issues early, such as missing fields or incorrect payload structures, before deployment.

It also made collaboration easier because clearly defined interfaces improved consistency across frontend and backend systems.”

4. What is REST API design? Can you explain how you've implemented or consumed APIs in your projects?

“REST API design is a standardized way of building communication between client and server systems using HTTP methods like GET, POST, PUT, PATCH, and DELETE.

A good REST API follows clear resource-based routing, proper request-response structures, validation, authentication, and meaningful status codes.

During my internship and personal projects, I implemented REST APIs using NestJS for handling workflows, entity management, authentication, and operational data processing.

I also documented APIs systematically using Swagger/OpenAPI, including request parameters, validations, response formats, and error handling, which improved collaboration and testing across the team.

On the frontend side, I consumed APIs in Next.js applications for dashboard workflows, trip analysis systems, and operational platforms.”

5. What is a process map, and why is it important in MES implementation projects?

A process map is a structured visual representation of how a business or operational workflow functions step by step, including activities, decisions, inputs, outputs, and responsible users or systems.

In MES implementation projects, process maps are important because they help understand how manufacturing operations currently work and how they should function after system implementation.

They help identify workflow dependencies, inefficiencies, validations, and operational requirements before configuring or implementing MES systems.

From my internship experience working on workflow-driven systems and documentation, I understand how important structured process understanding is for building reliable operational platforms.”

6. What does 'functional consultant' mean, and how does this role differ from a developer?

“A functional consultant focuses on understanding business processes, operational requirements, and workflows, and then translating them into system configurations, documentation, and functional solutions.

A developer mainly focuses on implementing the technical side through coding and system development.

The reason this role interests me is because it combines both technical understanding and operational problem solving.

My development background helps me understand systems, APIs, workflows, and backend logic, while I’m also interested in understanding how technology improves real-world business and manufacturing operations.

I think this combination can help bridge communication between technical teams and operational stakeholders effectively.”

7. How do you debug a technical issue you've never encountered before? Walk me through your process.

“When I encounter a technical issue I haven’t seen before, I first try to clearly reproduce and isolate the problem instead of making assumptions.

I carefully examine logs, error messages, request-response behavior, database data, and system flow to understand where the issue is occurring.

For backend issues, I often use tools like Postman, logging, and step-by-step validation of APIs and payloads. I then form hypotheses, test possible causes systematically, and narrow down the root issue.

During my internship, this structured debugging approach was very important while working on workflow-driven backend systems and scheduled cloud functions.

If needed, I also research documentation or discuss with senior team members rather than guessing and risking incorrect fixes.

I believe structured troubleshooting and calm analysis are very important in operational systems.”

8. What is Swagger, and how have you used it for API documentation?

“Swagger, based on the OpenAPI specification, is a tool used for creating interactive API documentation.

In NestJS, I used Swagger decorators to systematically document backend APIs, including endpoints, request payloads, query parameters, authentication requirements, validations, response formats, and possible error responses.

This documentation helped frontend developers, testers, and team members understand and test APIs more efficiently.

During my internship, I realized that clear API documentation is very important in workflow-oriented systems because it improves maintainability, collaboration, and troubleshooting.”

9. You mention N8N automation in your skills — what is it and how have you used it?

“N8N is a workflow automation platform used to connect different services, APIs, and systems through trigger-based workflows and automation logic.

I explored it primarily to understand how operational workflows and backend automations can be orchestrated visually and efficiently.

My interest in tools like N8N also connects with the kind of workflow-driven and automation-oriented systems I worked on during my internship, especially around scheduled backend tasks and process automation.

I’m particularly interested in how automation platforms help reduce manual operational effort and improve workflow consistency.”

10. If you don't know the answer to a technical question during a project, what do you do?

“If I don’t know the answer to a technical issue, I first acknowledge the gap honestly instead of making assumptions that could create larger problems later.

I then try to understand the problem systematically by checking documentation, analyzing logs or workflows, researching reliable sources, and experimenting carefully in a controlled way.

If the issue is impacting timelines or involves critical systems, I communicate clearly with senior team members and seek guidance when necessary rather than delaying silently.

During my internship, I learned that asking the right questions early and escalating appropriately is an important part of professional software development and operational reliability.

I believe a strong learning mindset and structured problem-solving approach are more important than pretending to know everything.”

